

3(a)	any five from: 1 <u>restriction endonuclease</u> ; 2 (RE) cuts, bacterial / herbicide-resistance, gene / DNA ; 3 (RE) cuts a, plasmid / vector ; 4 <u>DNA ligase</u> joins, new / bacterial / foreign, gene to, plasmid / vector ; 5 <i>ref. to</i> joining / adding, of <u>promoter</u> ; 6 (promoter / regulatory sequence) ensures expression of (new / bacterial / foreign / prokaryotic) gene ;	5
3(b)	73 / 100 × 123.5 or 90.155 million ; (90.155 × 2 =) 180.31 million ;	2

Question	Answer	Marks
3(c)	any four from: <i>positive implications:</i> 1 more food / higher yield (per hectare) ; 2 (farmers) use less, insecticide / pesticide ; 3 cheaper food prices for consumers ; 4 less land needed to grow food leaves more for, biodiversity / conservation ; 5 improved food nutritional value ; <i>negative implications:</i> 6 food allergies ; 7 greater quantity of herbicide residue on food ; 8 (evolution of) resistance in, insects / weeds / fungi / pests ; 9 pollen drift to / cross-breeding with, organic / non-GM / conventional, crops ; 10 GM crops may become invasive weeds ; 11 consumer mistrust / GM food must be clearly labelled ;	4

Question	Answer	Marks
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